

Recognize a 'typical' clinical presentation of MS

- MS is most common demyelinating disease of the CNS
- Younger, females higher risk
- Vision loss/Optic Neuritis, Sensory Abnormalities very common
- Heat Sensitivity
- Fatigue
- Remitting and relapsing symptoms
- Clinically distinct episodes with partial or total remission in between
- Smoking, CVD, Obesity are risk factors
- Lack of adequate health care access/cost may delay treatments/follow up

Clinical features of multiple sclerosis

Features suggestive of multiple sclerosis
Relapses and remissions
Onset between ages 15 and 50
Optic neuritis
Lhermitte sign
Internuclear ophthalmoplegia
Fatigue
Uhthoff phenomenon
Features not suggestive of multiple sclerosis
Steady progression
Onset before age 10 or after age 50
Cortical deficits such as aphasia, apraxia, alexia, or neglect
Rigidity, sustained dystonia
Convulsions
Early dementia
Deficit developing within minutes

UpToDate®

Recall environmental and genetic features that appear to be causally related to MS

- Potential environmental triggers/risks include lack of UVB light/Vitamin D
 - Living in latitudes with less sun exposure increase risk for MS. Moving from low sun exposure latitudes to latitudes with more sun exposure **before** puberty reduces risk.
- Epstein-Barr virus associated with MS; Other virus may potentially increase risk/trigger MS. (Lyme)
- Chromosome 6, MHC: HLA-DR2 class;
 - (DRB1*01:15 homozygotes 9x increased risk; Heterozygotes 3 x increased risk)
 - Concordance in identical twins 20-39%
 - Board Review mnemonic: Multiple Hay Pastures are Dirty referencing **Multiple** Sclerosis, **Hay** Fever and Good**pasture** Syndrome related to DR2 epitope.
 - FYI: Over 200 other genes with potential roles identified; epigenetics likely large role in manifestation of disease and response to DMT)
 -

Describe the pathophysiology and pathogenesis of MS and recognize associated pathology

- Genetic susceptibility and environmental trigger pathophysiology of MS
- T-cells activated in lymphatic tissue, enter the CNS creating inflammation and damage to myelin sheath. T cell signals activate B-Cells to create autoantibodies as well as continued stimulation of T-Cells and other cytokines. Immune activation targeted at myelin leads to gliosis, dendritic loss and myelin destruction.
- Loss of myelin exposes axons, affecting nerve transmission and ultimately leads to axon damage and degeneration.

State the major differences between MS and Guillain-Barre Syndrome

- MS is a disorder of myelin in the CNS; Guillain-Barre is a disorder of the PNS.
- MS is more frequently relapsing/remitting; GBS is rapid, ascending in nature.

List common neurologic and ancillary signs and symptoms of MS

- Progressive weakness with ataxic gait/gait disturbances
- Depression is associated with MS
- Spasticity with associated pain
- Bladder Dysfunction
- See features noted above.

Describe the 3 patterns of the clinical course of MS and identify the most common of the 3.

- First attack designated as Clinically Isolated Syndrome (CIS); Most will progress to the 3 core phenotypes of MS (Relapsing Remitting and Progressive, either primary or secondary)
 - 1. Relapsing Remitting MS (this is the most common, 85-90%)
 - Characterized in time by s/s that relapse and remit. There is *not* significant progression of disability in between the relapses.

- 2. Secondary Progressive: Begins as RRMS then ***secondarily*** progresses to secondary progressive MS; The patient may continue to have relapse and remissions of acute attacks however in progressive MS, the disability progresses. (they do not return to near normal baseline between attacks)
- 3. Primary Progressive: occurs in about 10% of cases.
 - Disability progresses from onset of diagnosis

Generate a differential Dx for MS and outline an appropriate work up, including understanding of the importance of MRI

Screen for “mimics”, other

- ANA (lupus)
- ESR (arteritis, other inflammatory/immune mediated disorders)
- Lyme Titers
- B12 Level
- ACE level (sarcoid)
- Other viral diseases, HIV
- ADEM, GBS
- See DDX list from Up to Date Below (you do not need to memorize every single disorder, but be able to come up with a ddx)

Differential diagnosis of multiple sclerosis

Inflammatory disease
Acute disseminated encephalomyelitis
Behçet disease
Chronic lymphocytic inflammation with pontine perivascular enhancement responsive to steroids
Clinically isolated syndromes suggestive of multiple sclerosis
Myelin oligodendrocyte glycoprotein IgG-associated encephalomyelitis
Neuromyelitis optica spectrum disorder
Paraneoplastic encephalomyelopathies
Polyarteritis nodosa
Primary angiitis of the central nervous system
Sjögren syndrome
Systemic lupus erythematosus
Infectious disease
Human immunodeficiency virus
Lyme neuroborreliosis
Neurosyphilis
Progressive multifocal leukoencephalopathy
HTLV-1-associated myelopathy/Tropical spastic paraparesis
Genetic disease
Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy
Granulomatous disease
Lymphomatoid granulomatosis
Sarcoidosis
Granulomatosis with polyangiitis (Wegener's)
Disease of myelin
Adrenoleukodystrophy
Adult metachromatic leukodystrophy
Other
Arnold-Chiari malformation
Compressive spinal cord lesions
Vascular malformations
Vitamin B12 deficiency
Spinocerebellar disorders

Human T-lymphotropic virus, type I; IgG: Immunoglobulin G.

UpToDate[®]

- MRI demonstrates ring enhancing lesions, typically ovoid shaped separated in space over time. MRI findings in >95% patients with MS. (lesions can be seen incidentally on MRI in patients without symptoms, always consider differentials)

Describe difference between treatments for acute relapses and disease modifying therapy and identify when each is used:

1. Acute Relapses: used to treat symptoms during an acute relapse
 - Shorten duration and severity of symptoms
 - IV (or oral depending on severity) high dose corticosteroids
 - For optic neuritis, always use IV
2. Disease Modifying Therapy (DMT): goal is to modify progression of disease
 - Recommended to start at time of diagnosis for RRMS (and associated SPMS)

- Some indications for PPMS, CIS (no consensus; use guidelines to aid decision)
- Evidence of reduction of relapses and MRI changes, possible reduction in severity; Does not reverse or cure the disease. Long term effects on morbidity and mortality *suggest* improvement.

Identify common acute relapse treatments and disease modifying agents for MS (for this MS lecture, you do not need to know specific drugs, dosages or detailed mechanisms of action; Recognize the general classes and very basic mechanisms of action, significant side effects of the classes of drugs; if drugs were presented elsewhere in this IS-NMSK module, learn them as per required by that faculty/lecture).

Acute Relapses:

- High Dose Corticosteroids (usually IV)
- Dalfampridine (see below)

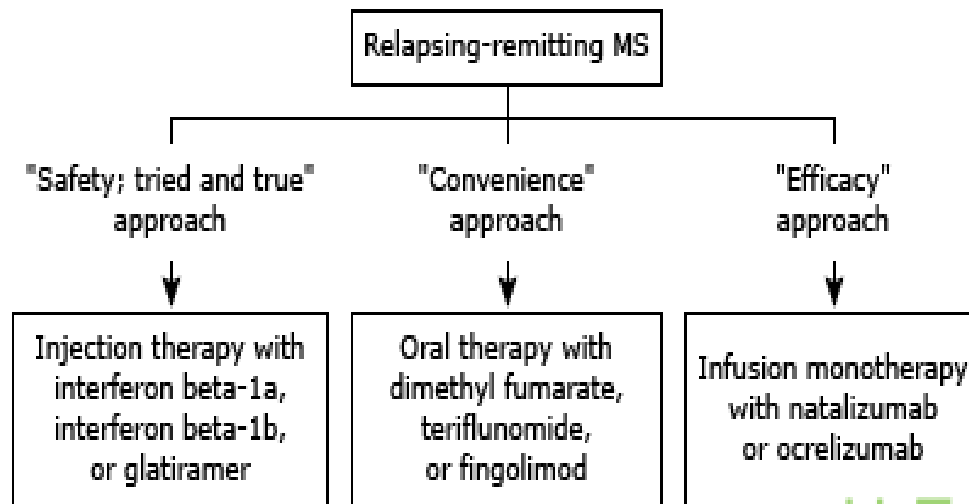
DMT

1. **Monoclonal Antibodies:** target and bind to receptors on immune/WBC's to prevent entry/destruction of myelin). Considered **most efficacy**; Injectable or IV only;
2. **Cladribine:** Purine Anti-metabolite; may have higher efficacy and is **oral**;
3. **S1P receptor modulators:** appear to sequester affect lymphocyte ability to enter CNS; intermediate efficacy; available **orally**;
4. **Fumarates:** mechanism uncertain; may active Nrf2, protecting against oxidative damage. intermediate efficacy; available **orally**. (phytonutrients in food such as green tea, onions, garlic, cruciferous vegetables like cabbage, Brussel sprouts may also stimulate Nrf2, don't forget importance of nutrition)
5. **Teriflunomide:** pyrimidine synthesis inhibitor; **oral**; avoid if planning pregnancy (NO DMT is approved for use in pregnancy; some, such as teriflunomide may stay in system up to 2 years after discontinuing)
6. **Beta Interferons:** immune signaling regulators; lower efficacy; **better safety profiles**.

Choice of drugs usually based on patient preferences after well informed discussion of risk/benefits/costs/etc. Approaches include considering preference for efficacy, convenience (oral) or safety profile as noted below:

Initial disease-modifying therapy for relapsing-remitting multiple sclerosis

Be aware of the main types of drugs for each of the 3 approaches (detailed knowledge of each drug in the class is not necessary; effect



UpToDate®

Describe proposed mechanism of action of dalfampridine and importance of testing for JC virus

- Dalfampridine has indication to help improve walking in patients with MS. It may also be used short term for acute relapses. Dalfampridine is a potassium channel blocker. It is theorized that destruction of myelin sheath exposes axons, disrupting the Na/K channels (excessive potassium) and ability to repolarize, thus affecting action potential. By blocking the potassium channels nerve conduction may be improved.
 - Increased temperature may also affect the ability of nerves to repolarize thus suggesting a mechanism of heat intolerance and increased symptoms (especially optical and sensory) in patients with MS.
- JC Virus (John Cunningham Virus, is a polyomavirus found in 50-70% of the healthy population)
 - JC Virus reactivation can cause **Progressive Multifocal Leukoencephalopathy (PML)**, a life-threatening disease.
 - It has been commonly associated with significant immune suppression in HIV/AIDS
 - With development of immune altering medications, including monoclonal antibodies, there has been increased risk for JCV activation and PML.
 - The DMT natalizumab has been associated with JCV and PML and thus **use of monoclonal antibody DMT should include screening for JCV before starting**

treatments and monitoring for signs and symptoms (which may be similar to MS relapse, although MRI demonstrates non-enhancing lesions).

List common treatments for ancillary symptoms (for this lecture, be able to consider choice of medications to reduce polypharmacy, costs and side effects as well as use of non-pharmaceutical interventions; You do not need to memorize specific dosages, side effects or mechanism of action for individual drugs. Be aware of general classes, basic mechanisms of action and side effects with-in classes of drugs. If you learned these drugs in other lectures you are responsible for the details as per that lecture/faculty).

Treatment for ancillary symptoms related to MS is important for quality of life. Common associated complaints include:

- **Gait impairment**
 - Dalfampridine = Ampyra® (extended release, oral, twice a day)
 - Potassium channel blocker possibly increasing conduction in demyelinated axons
 - OMT/Physical Therapy/Exercise Physiology/Occupational Therapy
 - Adaptive Equipment Evaluation (shoes, cane, walker, customized wheelchair, etc)
 - Social Support (walking buddy)
- **Spasticity**
 - GABA agonist (Baclofen = Lioresal®; oral, or in extreme cases, via implanted intrathecal pump)
 - Centrally acting α_2 adrenergic agonist (Tizanidine = Zanaflex®)
 - Long acting benzodiazepine (diazepam 2.5mg-10mg up to QID)
 - Dantrolene (25-100 mg up to QID)
 - Recognize most of these drugs SE include dizzy, weakness, drowsy, fatigue
typical symptoms already associated with MS; May increase risk for falls;
 - In some instances, reduction of spasticity may affect gait ability making it worse. Work with PT to help reduce these associated problems.
- **Pain and paroxysmal disorders**
 - Anticonvulsants (Gabapentin = Neurontin®, carbamazepine, phenytoin); Various classes and mechanism of action generally include side effects of drowsy, fatigue and possible neurological signs including ataxia and nystagmus.
 - Antidepressants (Tricyclics: amitriptyline, SSNI, Duloxetine): May help with pain, insomnia and offer anti-depressant effect. Older TCA's often less expensive. Monitor for side effects. (esp. anti-cholinergic effects)
 - Incorporate a Multi-faceted Pain Management Plan. (Treat the Whole Person!)
 - Pain Management Group, Behavioral Health Services, Social Work Services, OMT, PT, Nutrition Therapy, Biofeedback/Relaxation, Social Groups, acupuncture and other modalities)
- **Bladder urgency (neurogenic bladder)**
 - Anticholinergic Medications such as oxybutynin, tolterodine may be helpful with urge incontinence. History is important to determine mechanism of dysfunction. Urology consult may be beneficial.

Be aware of the various classes of drugs and what symptoms they would be used for.

You do not need to know each individual drug or dosages.

Be able to consider utilizing drugs that might provide more than one benefit. ie: if have pain and insomnia consider anti-depressant like a tricyclic

- Bladder training, education on fluid intake, avoiding caffeine and smoking cessation in addition to medication therapy.
- Avoidance of UTI; (may need prophylactic therapy if recurrent UTI)
- Urinary Catheterization may be required.
- **Fatigue:** Very Common
 - Rule out other contributors: anemia, thyroid problems
 - May be related to sleep disturbance-consider polysomnogram
 - Consider adding psychostimulants (methylphenidate = Ritalin®, modafinil = Provigil®) or amantadine
 - Diet/Exercise/Lifestyle (ask about trauma, stressful environment) REFER to SERVICES as part of your prescription.
 - Healthy diet to support energy production! Avoid processed, refined foods; Increase Vegetables and Fruits, whole grains; Essential Fatty Acids. If available, consultation with nutritionist educated in functional/precision nutrition.
- **Cognition**
 - Processing speed more affected than memory (tip of the tongue phenomenon is common)
 - Consider cognitive testing, especially with work-related problems
 - Consider long acting psychostimulant (long-acting methylphenidate formulation = Concerta®)
 - Think like a DO: consider structure and function, root causes and support the bodies innate ability to self-regulate and heal. Start with medications that target root cause as determined by thorough H&P; Diet and lifestyle modifications that target life stressors, environmental toxins, structural barriers, depression, sleep, exercise and nutrition may all benefit cognition.

Describe current opinions regarding pregnancy in women with MS

- MS affects women of child-bearing age. Discussion of reproductive issues is essential. A neurologist and OB should be involved in reproductive planning *before*, during and after pregnancy.
- MS does not appear to reduce fertility. MS symptoms may diminish during pregnancy. Symptoms may rebound/increase after pregnancy.
- DMT can be teratogenic. There are currently NO FDA approved DMT for use during pregnancy. (some neurologists use beta interferons;)
- DMT needs to be stopped prior to pregnancy. Some medications remain in system for extended periods after stopping. Proper contraception and timing essential.
- Rebound of symptoms after stopping monoclonal antibodies is a potential risk patient should be aware of before beginning the medication.

Describe basic features of ADEM: Acute Demyelinating Encephalomyelitis

- Acute multifocal CNS inflammation and demyelination typically after an acute infection and rarely immunization (5% cases occur after live measles vaccine: 1-2/1,000,000)
- Rapidly progressive symptoms including altered mental status; Often fever, HA, meningeal signs. (in contrast to MS)
- Frequently younger children
- Rx: high dose glucocorticoids

Recognize structural issues affecting treatment of MS particularly those in marginalized communities.

- Exceptional cost of DMT (>60,000-120,000/year)
 - Copay expensive even if have insurance; Drug pricing *policies* in US
 - Insurance company *policies* restricting use of expensive medications, requires difficult approval process.
 - Health Care Policies/Access to insurance in US
- Delayed presentation and treatment in marginalized communities
 - Research show LGBTQ+ populations less likely to seek prompt medical care due to previous discrimination while navigating health care system.
- Food swamps/deserts, environmental pollution, ACE's that may play role in MS and chronic illness.
- Policies in medical education that focus on disease and pharmaceutical management of symptoms with comparatively minimal focus on nutrition and lifestyle interventions.
- Bias in research protocols, mistrust in participating in medical research due to abuses and unethical practices as well as lack of resources lead to under representation of marginalized communities. FYI: The following sites are resources to share with patients to address discrepancies in research and improve opportunities.
- National African Americans with MS Registry (NAAMS): www.naamsr.org
 - Founded by neurologists, the goals of the NAAMS Registry are to have an accurate understanding of the number of African Americans afflicted with MS, their potential barriers to care, how this relates to where the patients live, and whether certain medications are of more benefit to African Americans.
- MS Minority Research Engagement Partnership Network: <https://www.acceleratedcure.org/about-network>
 - An inclusive and comprehensive program that brings together diverse stakeholder groups to identify and address the barriers to minority participation in MS research.
- CHIMES Trial: <https://chimesstudy.com/>
 - A multicenter study to assess disease activity and biomarker of neuronal damage in Black or African American and Hispanic/Latino patients with relapsing multiple sclerosis receiving treatment

Recognize osteopathic principles related to treatment of MS including the relationship of nutrition on structure and function and body's ability to self-regulate and heal.

OSTEOPATHIC PRINCIPLES

Person is Dynamic Unit of Mind, Body and Spirit

- What structural issues are effecting mind, body and spirit.
- ACE's* Chronic Stress, Minority Stress
- Environmental Factors/pollution; Food security (swamps, deserts, apartheid)

• Structure and Function are Reciprocally Inter-related at all levels

- What is the structure of the mitochondria? How can we enhance that?
- Somatoskeletal Structure: OMM to improve flexibility, myofascial (contains immune cells), reduce pain

• Body has Innate Ability to Heal Itself

• Rational Treatment Based on these Principles

- Be a DO

The diagram consists of four overlapping circles arranged in a square pattern. Each circle contains a principle of osteopathy. The top circle states: 'The body possesses self-regulatory mechanisms that are self-healing in nature'. The right circle states: 'The human being is a dynamic unit of function-mind, body and spirit'. The bottom circle states: 'Rational treatment is based on these principles'. The left circle states: 'Structure and function and interrelated at all levels'.

Obj: 13. Recognize structural issues affecting treatment of MS particularly those in marginalized communities.
Obj 1.4. Recognize osteopathic principles related to treatment of MS including the relationship of nutrition on structure and function and body's ability to self regulate and heal.